# Lesson 3 Face Recognition

## 1. Program Logic

Image recognition is widely applied in artificial intelligence and face recognition is the most popular, which is often used in door locks, smart phone face unlock, etc.

The overall implementation process of this lesson is as follow.

Firstly, use the face model trained before and zoom in or out the screen to recognize face. Then transfer the recognized face coordinate into the original coordinate to judge whether the face is the largest size and the face will be circled after recognition.

Next, make servo stabilizer rotate right or left to recognize face. By controlling the rotation angle of the bus servo, we can make the robot perform the feedback after recognition.

The source code is located in /home/ubuntu/armpi_fpv/src/face_detect/scripts.

```python
def run(img):
    global frame_pass
    global start_greet

    img_copy = img.copy()
    img_h, img_w = img.shape[:2]

    if frame_pass:
        frame_pass = False
        return img

    frame_pass = True

    blob = cv2.dnn.blobFromImage(img_copy, 1, (150, 150), [104, 117, 123], False, False)
    net.setInput(blob)
    detections = net.forward() #Computing recognition
    for i in range(detections.shape[2]):
        confidence = detections[0, 0, i, 2]
        if confidence > conf_threshold:
            #The recognized coordinates are converted to the coordinates before scaling
            x1 = int(detections[0, 0, i, 3] * img_w)
            y1 = int(detections[0, 0, i, 4] * img_h)
            x2 = int(detections[0, 0, i, 5] * img_w)
            y2 = int(detections[0, 0, i, 6] * img_h)
```

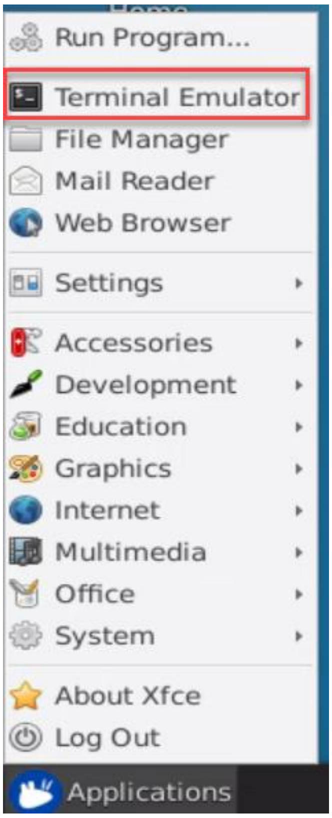
# 2. Operation Steps

The entered command must be strictly distinguished between uppercase and lowercase and spaces, and the keywords support the "TAB" key completing.

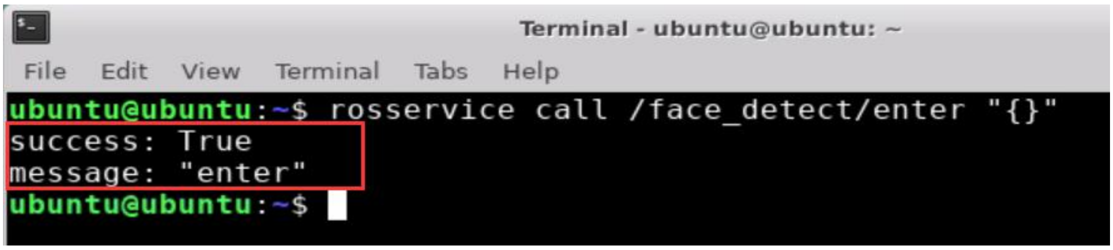
## 2.1 Enter the game

① Turn on ArmPi FPV and connect to Raspberry Pi desktop through NoMachine.

② Click “Applications” button and select “ Terminal Emulator” in pop-up interface to open the command line terminal.



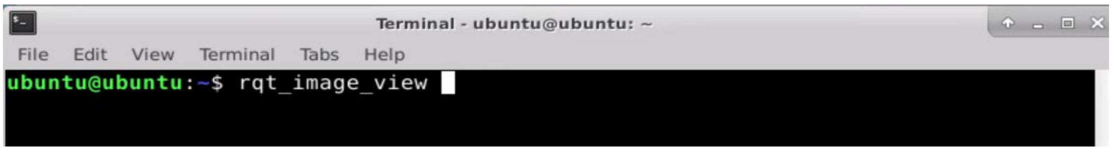

③ Enter “rosservice call /face_detect/enter "{}"" command, then press “Enter” to start the game. After the game starts, a print prompt will appear, as shown in the figure below.



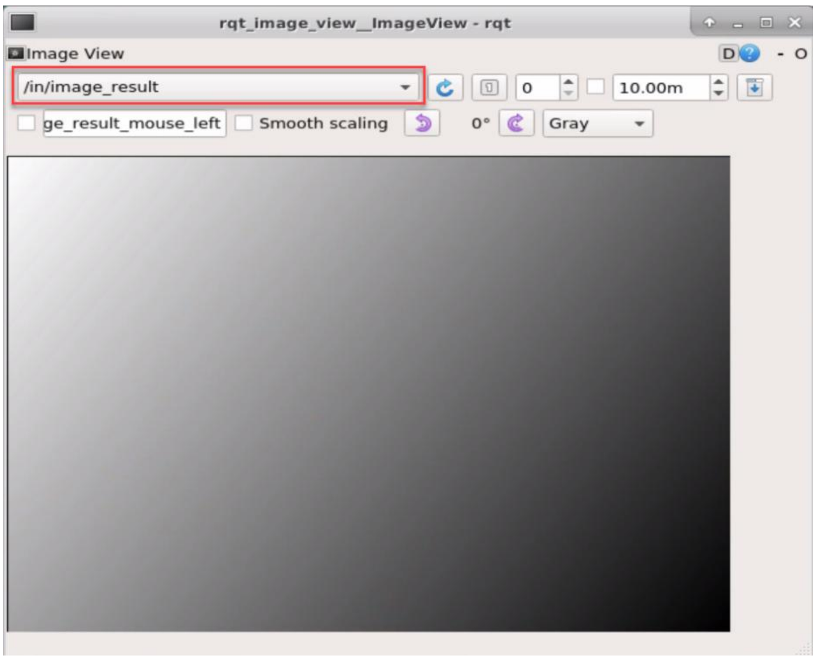

## 2.2 Turn On Image

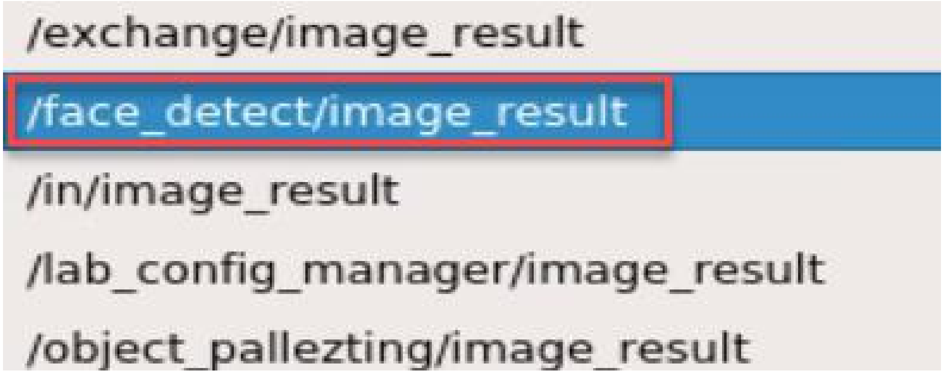
① When entering the game, we need to turn on image returned by camera with rqt. Please go back to the desktop and click the “Application” icon at bottom-left corner to open a new terminal.

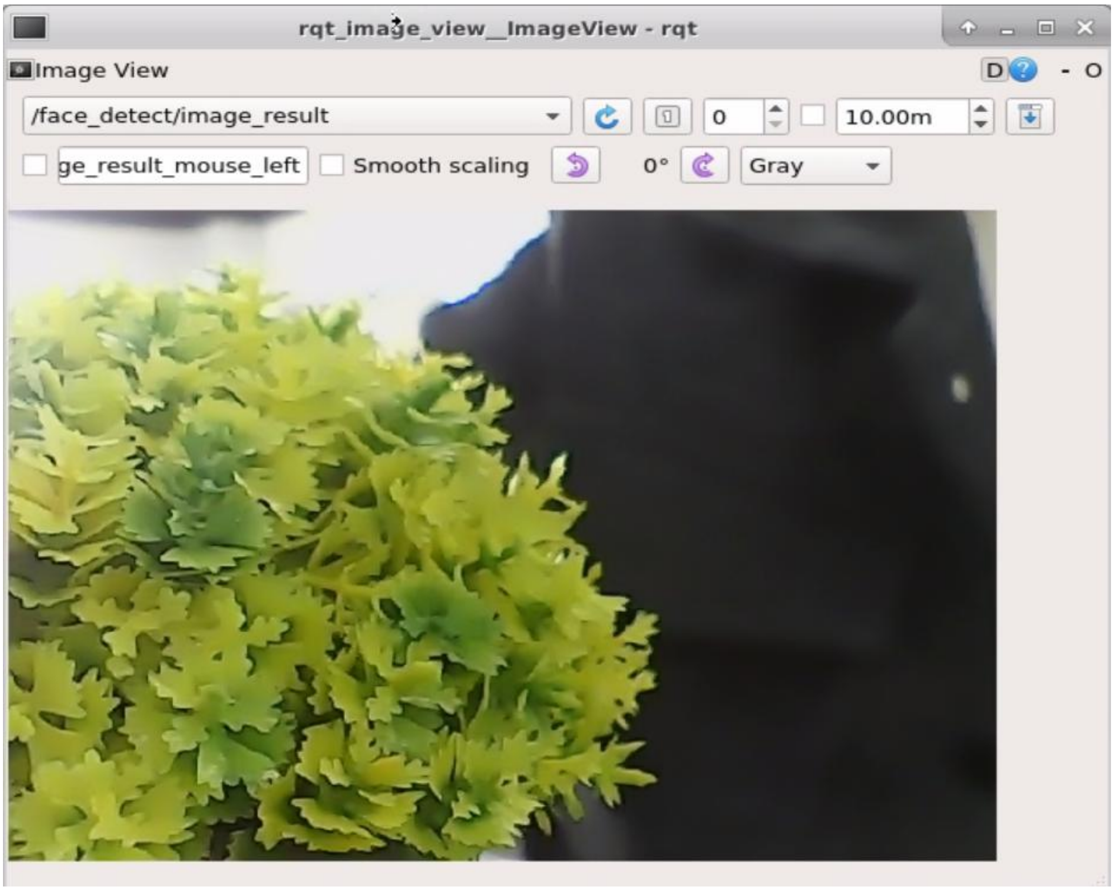
② Enter “rqt_image_view” command and press “Enter”. Then open rqt after few seconds.




③ As shown in the following figure, select “/object_tracking/image_result” and other settings remain the same.






Note: after the image is turned on, please be sure to select the topic corresponding to the game you play. Otherwise, the recognition process can not be displayed normally if you start other games.

## 2.3 Start the Game

Please go back to the terminal opened in “2.1 Enter the Game” and enter “rosservice call /face_detect/set_running "data: true"” command. When the prompt, shown in the red box, appears, it means you start the game successfully.

```
ubuntu@ubuntu:~$ rosservice call /face_detect/set_running "data: true"
success: True
message: "set_running"
ubuntu@ubuntu:~$
```

## 2.4 Pause and Quit the Game

If you want to pause the game, you can enter “rosservice call /face_detect/set_running "data: false"" command.

```
ubuntu@ubuntu:~$ rosservice call /face_detect/set_running "data: false"
success: True
message: "set_running"
ubuntu@ubuntu:~$
```

For quitting the game, you need to enter “rosservice call /face_detect/exit "{}"" command.

```
ubuntu@ubuntu:~$ rosservice call /face_detect/exit "{}"
success: True
message: "exit"
ubuntu@ubuntu:~$
```

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Note: the current game will continue as Raspberry Pi is powered on. To avoid occupying too much RAM, please quit the current game according to the instruction above before playing other AI vision game.

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If you want to turn off the image returned by camera, you can turn back to open rqt terminal and press “Ctrl+C”.

## 3. Function Realization

When the game starts, the robotic arm will rotate left and right. After the face is recognized, the target face will be framed. At this time, the gripper of the robotic arm will rotate back and forth.